

Abirami Valliammai Kathiresan

akathiresan@ucdavis.edu | 408-217-8537 | <https://www.linkedin.com/in/abirami-kathiresan> | <https://abiramikathiresan.vercel.app/>

Education:

University of California, Davis | Davis, CA

September 2023 - June 2027

- B.S. in Data Science, Minor in Computer Science & Tech Management | GPA: 3.98, Deans' Honor List (Fall 2023 - Spring 2025)
- Relevant Coursework: Data Structures in Python, Data Structures in C++, Databases, Machine Learning, Algorithms for Data Science, Discrete Mathematics, Statistics for Data Analytics, Calculus, Linear Algebra

Experience:

Software Engineering Intern | Charles Schwab

June 2026 - Current

- Migrating 5+ infrastructure automation playbooks with Ansible and YAML for an internal tool monitoring application and service health, streamlining reliability workflows to support scalable operations across Schwab's container ecosystem.
- Developing 10+ REST APIs with Java to support automated diagnostics and health validation for enterprise applications and services.
- Building and deploying AI agents to streamline operational workflows and surface diagnostic insights for the Container Platform Reliability team, leveraging LLM-powered automation to improve engineering productivity.

Software Developer | ASUCD Innovation & Research Lab

January 2026 - Current

- Developing and maintaining the UC Davis mobile application using React Native, TypeScript, Expo, SQLite, and REST APIs in collaboration with the student government, undergoing beta testing before deployment to 40,000+ UC Davis students in Fall 2026.
- Engineered library room reservation system with optimistic locking, while using SQLite and Drizzle to mitigate third-party API delays.
- Debugged and optimized real-time transit prediction workflows across 20+ bus routes, improving arrival-time accuracy and reliability.

Software Engineering Intern | Equitas Bank

June 2025 - August 2025

- Developed 7+ REST APIs using Java and Spring Boot to support key internet banking operations, shipping code quickly and achieving average response times below 200 ms while collaborating with team members to iterate based on feedback.
- Built a backend-for-front-end (BFF) layer to streamline communication between microservices and the frontend, reducing API latency by 30% for thousands of daily users, and deployed services on AWS to ensure scalability in a fast-paced environment.
- Implemented and validated SQL queries on Oracle databases using Postman for 50+ tests, demonstrating strong problem-solving skills and technical collaboration.

Product Management Fellow | Product Space

October 2024 - Current

- Participated in a fellowship program to design and prototype Instagram Scraps, a feature that enabled Instagram users to customize their profiles, showcasing an ability to tackle end-to-end engineering challenges and collaborate cross-functionally.
- Leveraged interdisciplinary skills in product management to design and engineer a user-centric solution and developed interactive wireframes and prototypes in Figma, integrating direct user feedback to enhance interface clarity by 40%.

Full-Stack Developer | UC Davis #include

October 2024 - March 2025

- Designed and developed a responsive web application for the Joan Viteri Memorial Clinic using React, Node.js, HTML, CSS, and JavaScript, supporting 1000+ daily active users, ensuring fast iteration cycles and effective API integration.
- Implemented modern web development practices, including dynamic content rendering and responsive design, which increased mobile engagement by 30% and reduced load times by 25% in a collaborative, agile environment.

Projects:

ANCOR

- Developed a full-stack political literacy platform using React, Node.js, Firebase, and web scraping, to aggregate political news, ballot propositions, and educational resources and make government information more accessible to voters.
- Implemented scalable backend services and responsive UI components, enabling real-time delivery of 200+ political terms, legislative updates, and election-related articles across desktop and mobile devices.

RiskPredictor

- Designed a machine learning system to help international students assess deportation risk by combining demographic data with trends from 200+ web scraped immigration-related news articles.
- Trained classification models to 92% accuracy with preprocessing and feature selection, and deployed a Streamlit interface for real-time, visualization-supported predictions.

Technical Skills:

- Languages: Java, Python, JavaScript, TypeScript, HTML, CSS, C++, R, SQL, Bash, YAML
- Frameworks and Libraries: Spring Boot, React, Node.js, Express.js, BrainFlow, MNE, scikit-learn, PyTorch, NumPy, Matplotlib, Spark
- Tools: Git, AWS, Linux, Docker, Gradle, Maven, Jira, Postman, Figma, Streamlit, Oracle, MySQL, PostgreSQL, MongoDB, PCF